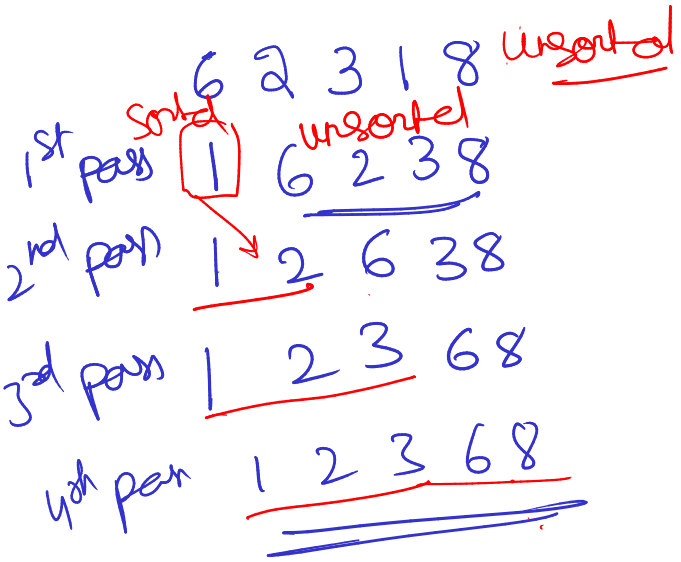
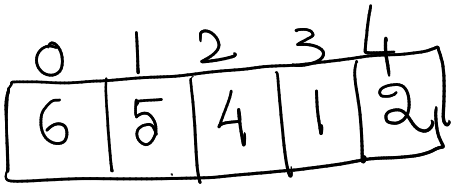
Selection Sort

1. Selecting smallest element in every pass
2. first array will be unsorted
3. combination of Sorted & unSorted
4. Until my array is completly Sorted





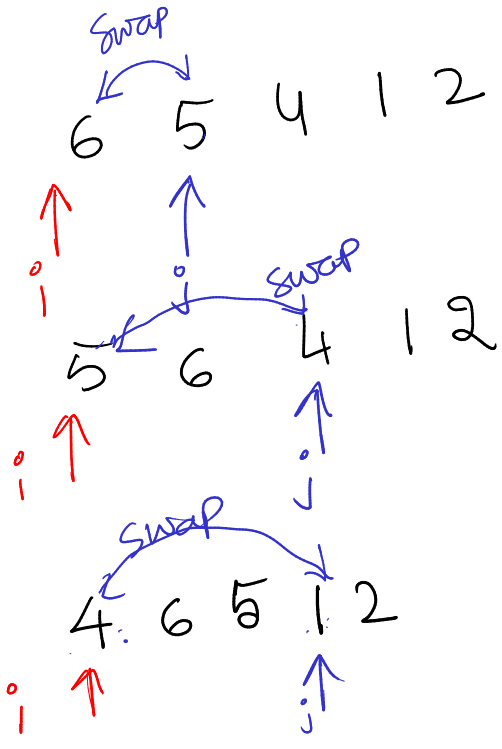
2 loops

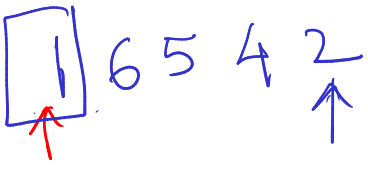
1st loop → i

2nd loop → j

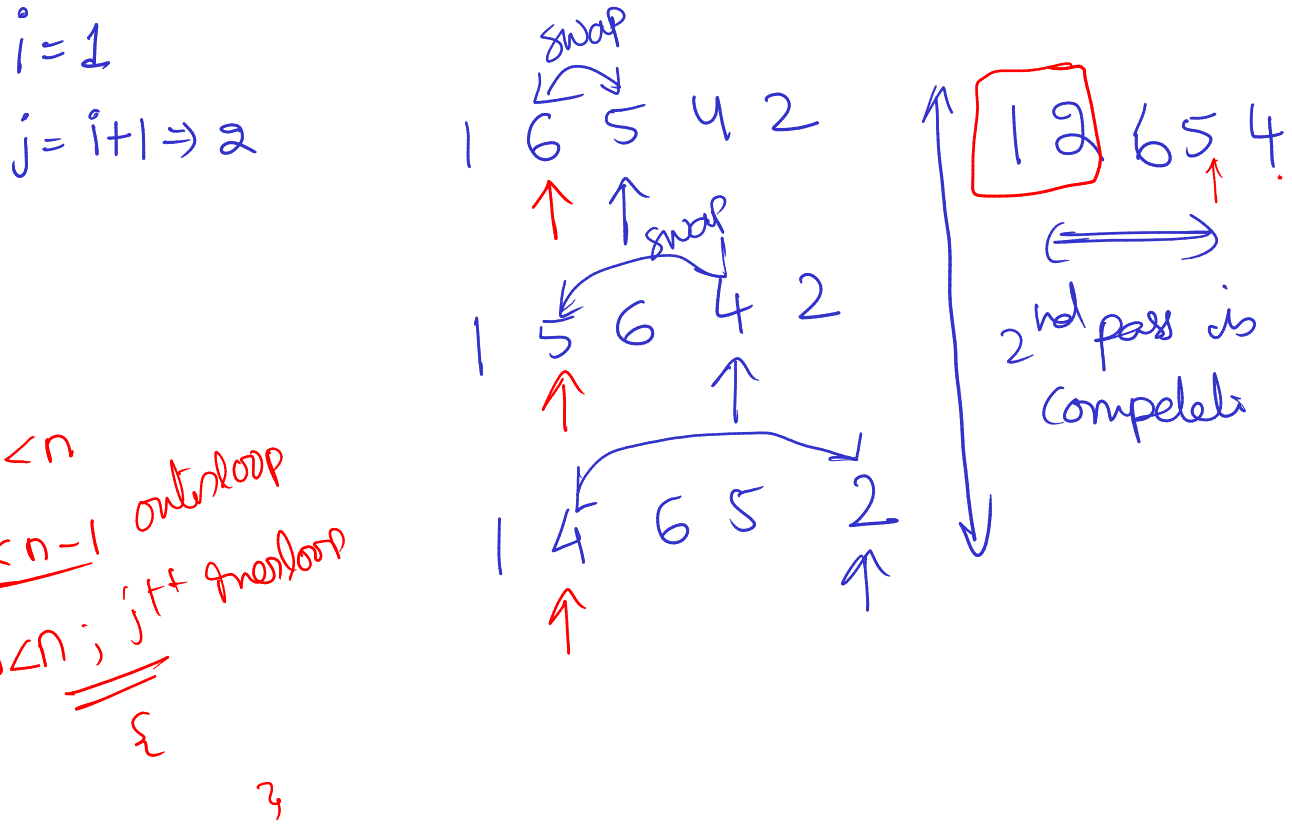
i = 0

j = i+1





1st pass is compeleted




```
i<n
i<=n-1  outerloop
j=i+1, j<n; j++ innerloop
{
}
```

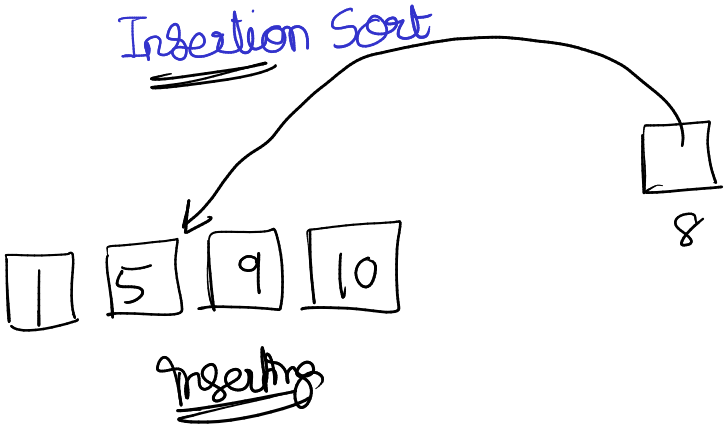
1. First Select an element by element from the array
2. compare that with all the elements and find out the smallest ele & place it in its appropriate position.

TO find Smallest element for each pass:

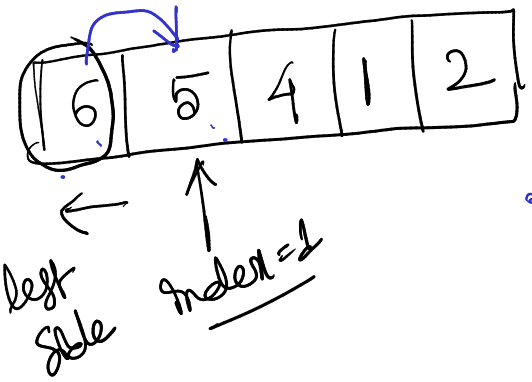
if the selected element is Smaller with the element in their respective pass swap the elements till the pass is completed

3. Repeat the above 2 steps until you get sorted array.

Insertion Sort



<=n-1
<n



i=1 a[i]=5

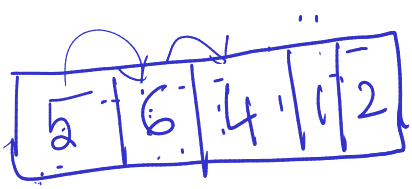
Key=5

j=i-1

j=0

if a[j]>key (or) key<a[j]

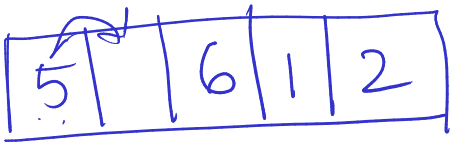
j is terminated -1





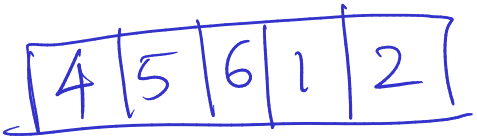
i=2 → a[i]=4 Key=4

j=1



4<a[1]

4<6

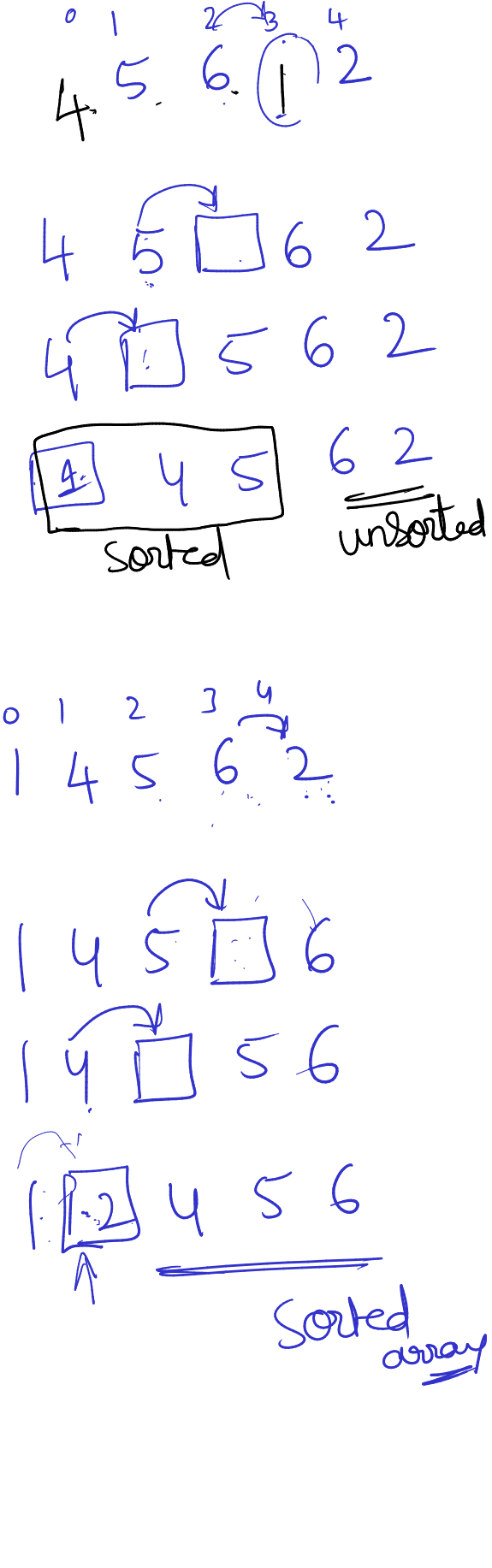


j=0

j=-1

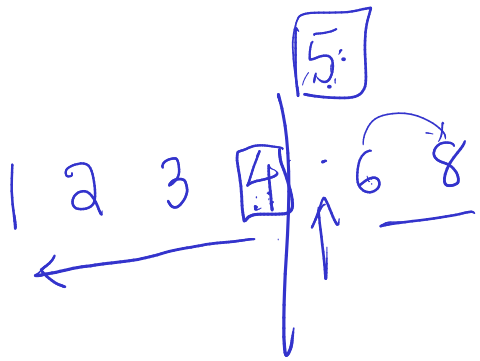
key<a[j]

4<5



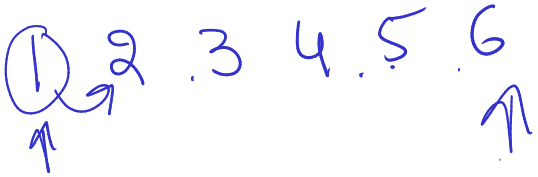
1. Always first element is considered as sorted element

2. Select the key element in the array
3. compare with the left elements present to it & insert the element at its appropriate position by comparing key element with the elements present to its left side.
4. Repeat the processes until the array is sorted.

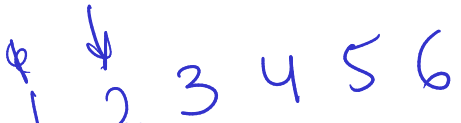


Time complexity

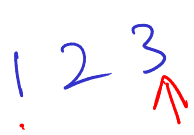
	Selection	Insertion
1. best case :	$O(n^2)$	$O(n)$
2. average case :	$O(n^2)$	$O(n^2)$
3. worst case :	$O(n^2)$	$O(n^2)$



sele



ins



i/p of string

1. scanf ⇒ scanf("%s", s)
2. gets
3. fgets

Instead of "%s", s
I write
"%[^\n]s", s

fgets(s, 10, stdin);

char

vector 6 ↕ vector in 7